

Market Guide for Telemetry Pipelines

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Initiatives: Delivery of Functional Responsibilities; Transform Cloud Strategy for Disruptive Platform Demands

Telemetry pipelines empower I&O leaders to manage the exponential growth of observability data, resulting in significant cost optimizations. This document will help to navigate the maturing telemetry pipeline market, as organizations seek to better manage their growing data streams.

Overview

Key Findings

- The distributed nature of workloads has resulted in the proliferation of telemetry sources, leading to inconsistent and nonstandard data formats that make analysis challenging. The traditional approach of indiscriminate ingestion is no longer effective.
- Telemetry pipelines provide a centralized solution for managing observability data by enabling the transformation, enrichment and routing of telemetry across both centralized and distributed environments, ensuring secure and scalable data transport.
- Most telemetry pipeline tools support both I&O and security operations use cases; however, recently, multiple telemetry pipelines have emerged that are centered on security operations.

Recommendations

- Improve telemetry pipeline management and value by planning for the impact on performance, cost and compliance while normalizing and routing data to chosen destinations.
- Optimize the flow, analysis and retention of telemetry by understanding and centralizing your data. Enhance your multicloud management by establishing a standard taxonomy that simplifies analysis and reinforces knowledge of your environment.
- Weigh consistency against other priorities when choosing telemetry pipeline tools. In some cases, security operations data requires unique handling, and incorporating SecOps-centric capabilities can bolster the business case.

Strategic Planning Assumption

Projections suggest that by 2027, 40% of all log telemetry will be processed through a telemetry pipeline product, a substantial increase from less than 20% in 2024.

Market Definition

Gartner defines telemetry pipelines as tools that provide a uniform and holistic mechanism to manage the collection, ingestion, transformation and routing of IT operational telemetry from source to destination. They can be used to filter and manage the amount of telemetry ingested by aggregation and analysis solutions, but have uses beyond cost management, such as data normalization and enrichment. As the number of telemetry sources increase and are distributed geographically, telemetry pipelines are becoming the core of an organization's telemetry life cycle management strategy.

The telemetry generated by today's digital platforms is ever increasing in volume and complexity. It must be rapidly ingested by analysis systems to ensure that alerts, events and exceptions are generated in a timely manner to notify operations and security teams and ensure that these workloads continue to meet performance and availability requirements.

Unfortunately, considerations for managing telemetry flow, analysis and life cycle are often overlooked as workload design and placement decisions are made. In addition, given the diversity of telemetry generators and analysis systems in the modern enterprise, inconsistency can translate to inefficient analysis and inconsistent results.

Example use-case scenarios or business problems addressed by telemetry pipelines include:

- Managing the cost of monitoring and observability tools by curating the amount of data ingested.
- Establishing and enforcing a uniform taxonomy for tags and identifiers, which improves telemetry quality and simplifies analysis and event generation by reducing variability.
- Routing telemetry to destination(s) based on the policy framework, sender or on the contents of the telemetry data itself.
- Masking PII or sensitive information in telemetry in a single location, rather than configuring it in every analysis system.
- Supporting both operational and security constituencies with the same data, reducing redundant sources of truth.

Mandatory Features

At a minimum, telemetry pipelines must:

- **Ingest** metric, event, log and trace (MELT) telemetry natively by API and/or software agent.
- **Integrate** with source or destination systems using industry standard or well-known vendor interfaces such as syslog, Splunk HEC and OpenTelemetry OTLP.
- **Transform**, filter and reformat data flowing through the pipeline per customer specification. This can be driven using regular expressions or special purpose languages.
- **Enrich** data as it passes through the pipeline using one or more external data sources, such as geolocation.
- **Route** data to one or more destinations based on data content and/or data source.

Common Features

Telemetry pipelines commonly include additional capabilities, such as:

- **Replay:** Although telemetry pipelines do not act as permanent repositories, the ability to replay recently processed data can be useful for testing or correcting transformation or enrichment errors.
- **Anomaly detection:** Identifying changes to or anomalies in volume and content can provide an early warning of behavior changes that may indicate problems.
- **Analysis:** Some telemetry pipelines have the ability to perform complex analysis on the data flowing through them, beyond what is necessary for transformation and enrichment.
- **Event generation:** When anomalies are detected or analysis generates notable results, events or alerts may be generated to notify teams.
- **Visual design surface:** A graphical user interface that allows users to create, modify and test pipelines' would work.
- **Automation:** Support for automation tools such as Ansible or Terraform, along with a "pipelines as code" capability allows telemetry pipeline tools to fit into the automated toolchains in use in many organizations.
- **Support for additional telemetry types:** Beyond MELT, some telemetry pipelines include support for profiles, network flows and packets, or other forms of telemetry. Support for data in OpenTelemetry format is common.

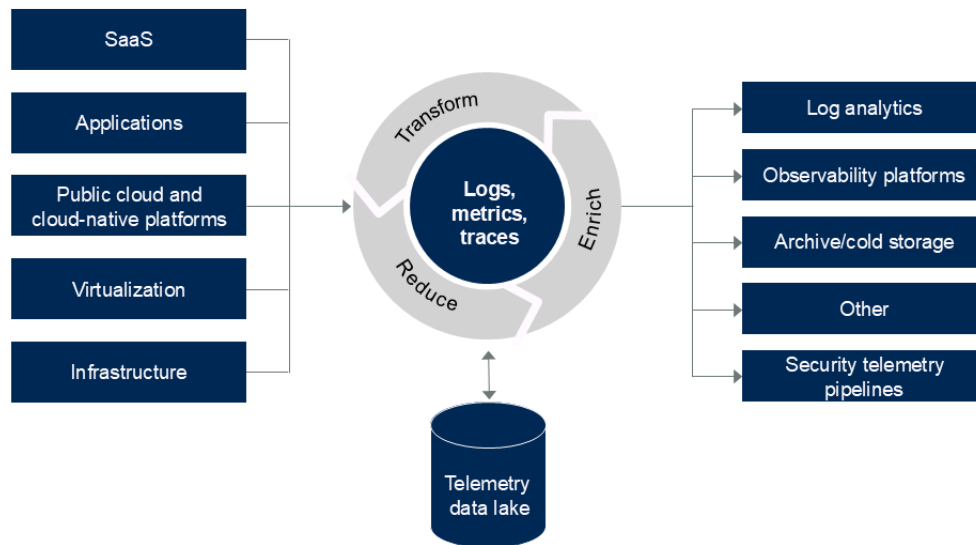
Market Description

Telemetry pipelines have emerged as an intermediate solution or a focal point to manage the collection, enrichment, transformation and routing of operational data from sources to destinations. These tools provide a uniform and holistic mechanism for this purpose. Historically, the burden of manipulating telemetry fell solely on the system of analysis, such as Elastic or Splunk. Manual configuration for managing telemetry was inefficient and failed to scale as telemetry increased, contributing to issues like false positives, duplicate data, extraneous data and unstructured data, which limited proactivity. Traditional data engineering products often could not support the low latency, volume elasticity and compatibility required for handling health, performance and security telemetry.

Telemetry pipelines address these challenges by taking telemetry from multiple data sources, unifying it and allowing it to be enhanced, transformed, reduced and routed to various destinations for analysis, storage, alerting and other purposes (see Figure 1). They offer capabilities to transform, redirect, enrich and reduce telemetry before it reaches its destination.

Figure 1: Telemetry Pipeline Schematic

Telemetry Pipeline Schematic



Source: Gartner
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Key Benefits of Implementing Telemetry Pipelines

- **Cost management and volume reduction.** This is the main benefit of telemetry pipeline solutions and, with the features it has, it can reduce the volume of metrics, events, logs and traces.
- **Flexibility with how telemetry is collected.** A telemetry pipeline can connect to and receive telemetry from various agents (proprietary and open-source) reducing the need to deploy its own agent at the source, although this option is available.
- **Data taxonomy establishment and reinforcement.** The benefit of this is standardized telemetry that standardizes fields and adds additional context, making telemetry more valuable to data consumers.

- **Eliminate vendor lock-in.** The benefit of this is allowing flexibility on the destination side of the telemetry pipeline. It eliminates proprietary ecosystem lock-in and enables a multitool strategy where telemetry can be sent to separate destinations such as observability tools, security information and event management (SIEM), cost-effective storage and other destinations.
- **Data protection and privacy.** The benefit of this is centralizing the protection of data in motion before it arrives at the destination. With centrally managed data protection in the pipeline, this takes the burden off destination systems, allowing them to dedicate system resources to their main functions.
- **Telemetry consolidation and standardization.** The benefit of this is to break down silos by unifying telemetry into a single pipeline allowing for centralized management of all telemetry in the enterprise. This, in turn, relieves burdens on system resources, engineering resources and organizational budgets.

Telemetry Pipeline Implementation Risks

- **Additional infrastructure or new products require additional support.** Organizations considering telemetry pipeline software are adding new infrastructure into their environment, which will require additional resources for configuration and maximizing the value of the solution.
- **Cost savings are not guaranteed.** Savings require a well-planned strategy and proper configuration. This includes managing data volume, selecting appropriate routing destinations, transforming and optimizing data, and minimizing the operational effort needed.
- **Skill and knowledge gaps.** Since this product technology is still new, organizations need training and hands-on experience to use it effectively.
- **Latency with alert generation and notification.** This is due to additional routing and increased travel time. Additionally, back pressure can develop when data is being ingested faster than the rate it can be processed and sent to its final destination.

Market Direction

The telemetry pipeline market is experiencing rapid evolution, driven by the increasing volume and complexity of operational data in modern IT environments. This market was nonexistent before 2018 but grew to include approximately six to 10 dedicated product vendors by 2025. Gartner's client inquiries regarding telemetry pipelines have increased substantially between 2023 and 2025, indicating a significant rise in adoption and interest.

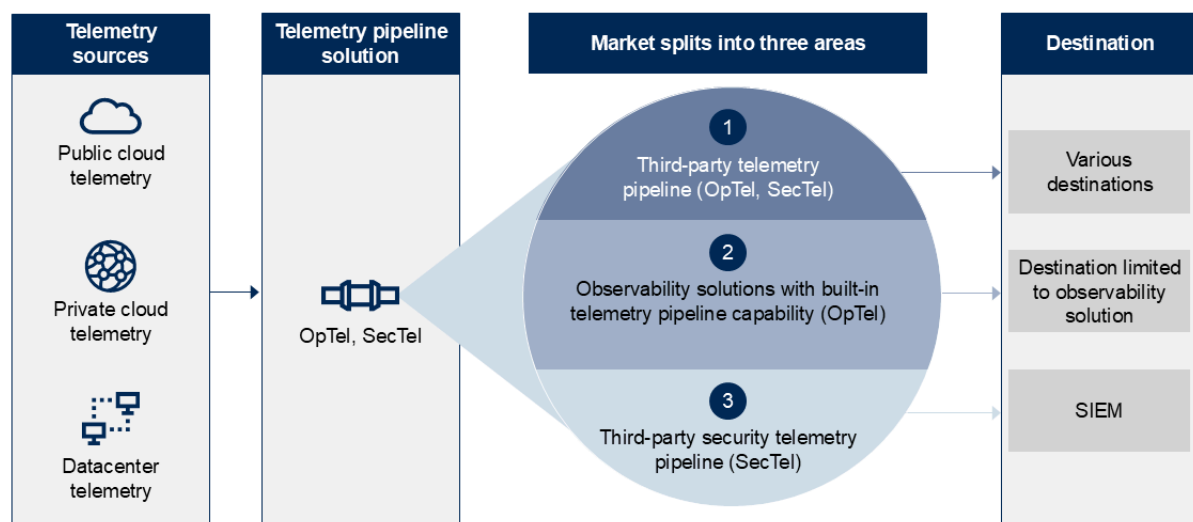
In the year ahead, the market for telemetry pipelines will continue to evolve, with several key trends and factors influencing its direction:

- **Increased adoption for cost management.** A primary driver for organizations adopting telemetry pipelines is the desire to manage the soaring costs associated with ingesting large volumes of telemetry (logs in particular) into expensive analysis systems. As telemetry volumes continue to grow, solutions that effectively filter unnecessary data and route noncritical information to lower-cost storage will gain significant traction.
- **Emphasis on data quality and standardization.** The proliferation of telemetry sources has led to inconsistent and nonstandard data formats, making analysis challenging. Telemetry pipelines, with their transformation and enrichment capabilities, will become increasingly crucial for establishing and enforcing a uniform taxonomy for tags, identifiers and attributes; thereby improving data quality and simplifying analysis. The ability to convert unstructured text logs into structured formats like JavaScript Object Notation (JSON), or extract metrics from logs, will be a key capability.
- **Broader integration and vendor neutrality.** Organizations often face vendor lock-in due to telemetry data being maintained in silos, as well as proprietary agent technology from vendors. The market will trend toward solutions that offer strong integration capabilities with various sources — including proprietary agent technologies — and support multiple destinations, such as observability platforms, SIEM systems, time series databases and cost-effective storage options. The ability to route data based on content or source will become increasingly important for flexibility and breaking down silos.

As the market evolves, vendors of observability platforms have adapted their strategies to address competition from independently developed third-party telemetry pipeline vendors. In addition to the telemetry market evolving, the observability market is now starting to include telemetry pipeline capabilities within their observability solution offerings. Another trend is the rise of security telemetry pipelines to handle security telemetry (SecTel) data. This does not leave third-party telemetry pipelines out of the picture entirely; rather, it adds capabilities to ingest and process SecTel data in addition to operational telemetry (OpTel), which several telemetry pipeline vendors are already doing. See Figure 2 for a depiction on these evolutionary trends:

Figure 2: Evolutionary Trends of the Telemetry Pipeline Market

Evolutionary Trends of the Telemetry Pipeline Market



Notes: OpTel = operational telemetry, SecTel = security telemetry
Source: Gartner
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Further key trends for the telemetry pipelines market include:

- **OpenTelemetry as a standard.** OpenTelemetry (OTel) has emerged as the open standard for distributed tracing, and has gained significant industry support. Telemetry pipeline products that support OTel-formatted telemetry and can transform non-OTel-compatible data to this standard will see increased demand. This will enable organizations to distribute OTel data to multiple destinations and leverage more sophisticated sampling criteria than typically available in OTel collectors.

- **Advanced capabilities beyond basic routing.** While cost reduction is a primary driver, organizations will increasingly look to leverage the additional capabilities of telemetry pipelines, such as on-demand replay of recent data for testing or error correction, anomaly detection for early warnings and complex in-pipeline analysis. The ability to generate events or alerts directly from the pipeline for immediate action will also gain importance.
- **Automation and “pipelines as code.”** As IT operations become more automated, telemetry pipelines will need to integrate seamlessly with existing automation toolchains — supporting “pipelines as code” capabilities. This aligns with broader DevOps practices that emphasize full life cycle automation, and treating infrastructure and configuration as code.
- **Skill gaps and infrastructure considerations:** Despite the benefits, the relatively new nature of dedicated telemetry pipeline products means that skill and knowledge gaps are present within organizations. The market will need to provide better training and simpler configuration mechanisms (e.g., visual design surfaces or automated processes) to facilitate adoption. Additionally, organizations will continue to grapple with the need for additional infrastructure to host and scale these solutions, whether self-managed or SaaS-based.
- **Strategic partnerships with telemetry pipeline Vendors:** Through strategic partnerships, additional telemetry outside of metric, event, log and trace (MELT) — such as network packet brokers — can now be sent to the pipeline. The intention of this to address gaps left due to rudimentary coverage of, for example, network monitoring within observability solutions.

The market direction is clearly trending toward telemetry pipelines becoming a central component of an organization’s telemetry life cycle management strategy — providing granular control, cost optimization and enhanced data utility across diverse IT environments.

Market Analysis

Managing the surge of IT operational telemetry has become a crucial challenge for organizations today. Modern environments — including cloud environments, hybrid setups and multicloud platforms — produce large volumes of machine data. This data primarily consists of metrics, events, logs and traces (MELT), along with other types of telemetry. The sheer volume of machine data generated can easily reach hundreds of terabytes (or even petabytes), leading to significant cost increases and expanded operational complexities. Large enterprises can spend over \$10 million annually just managing this data.

A core issue is the fragmentation and inconsistency of telemetry data resulting from organizational silos and the use of disparate tools, often with different agents. This creates redundant data collection and transmission, limits flexibility in routing data to different destinations, hinders unification and enrichment, and complicates the maintenance of consistent data governance policies. The traditional approach of manual configuration to manage data flow is inefficient and cannot scale with the increasing volume and complexity.

Telemetry pipelines have emerged as a direct response to these challenges. They serve as an intermediate layer providing a uniform and holistic mechanism to manage the flow of operational data from diverse sources to multiple destinations. These destinations can include SIEM systems, log analysis platforms, object storage, time series databases, application performance monitoring (APM) and network performance monitoring (NPM) tools. Commercial products have appeared on the market to simplify the implementation of this approach.

Solutions in this market offer several key capabilities that are essential for effective telemetry management:

- **Collection and ingestion:** The ability to gather telemetry natively via API or software agent from various sources.
- **Transformation:** Modifying data formats or structures to be consumable by destination systems. This is crucial for data normalization and can include reformatting or masking sensitive data like personally identifiable information (PII) or payment card industry (PCI) information.
- **Enrichment:** Adding context or attributes to the data, such as IP or geolocation, to increase its value for analysis.

- **Reduction and filtering:** Decreasing data volume by filtering out extraneous, duplicate or low-value data. This directly addresses the challenge of high data ingestion costs into volume-based analysis tools and helps clean up telemetry.
- **Routing:** Directing processed data to one or more destinations based on predefined policies or the data's content. This enables strategies like sending less critical data to lower-cost storage.

The adoption of telemetry pipelines is growing significantly. Even though cost reduction is a major initial driver, organizations are increasingly recognizing and leveraging other capabilities to maximize the value of their telemetry.

Implementing a telemetry pipeline represents an intentional strategy for managing the entire telemetry life cycle. Beyond cost savings, these solutions enable consistency in naming and taxonomy to simplify analysis, improve collaboration among operations teams by reducing data silos, and support security and compliance through centralized data masking. The efficiency gained in processing telemetry can also lead to reductions in mean time to detect (MTTD) and mean time to repair (MTTR) by decreasing false positives, duplicate events and alert noise.

However, organizations considering telemetry pipelines should be aware of potential challenges. These solutions are still relatively new, which can lead to skill and knowledge gaps within teams. There are also additional costs associated with acquiring and maintaining a separate solution, whether SaaS-based or self-hosted. Implementing the required infrastructure and ensuring compatibility with a wide range of existing sources and destinations can also pose difficulties. While ROI from cost savings is clear, demonstrating the return on investment for other benefits may require more justification. The market is also somewhat divided, with some vendors focusing on specific areas like cybersecurity. Despite potential latency concerns due to the added layer, pipeline solutions often include features like buffering and sampling to handle data spikes and back pressure effectively.

Telemetry pipelines can be deployed on-premises, as SaaS or in hybrid models. They can be stand-alone offerings or part of a vendor's broader monitoring portfolio. See Table 1 for the representative vendors in this market. These examples illustrate the types of offerings available but do not constitute an exhaustive list.

Representative Vendors

The vendors listed in this Market Guide do not imply an exhaustive list. This section is intended to provide more understanding of the market and its offerings.

Vendor Selection

The telemetry pipeline market is maturing as organizations seek to better manage their growing data streams. These solutions centralize telemetry management, providing tools to **transform, filter, enrich and route data**, making it more valuable for analysis and improving operational efficiency. To be successful, implementing these pipelines requires careful planning, updated data governance and a focus on using all their capabilities, not just cutting costs.

Available from various vendors as stand-alone products or as part of broader observability platforms, telemetry pipelines typically support SaaS, self-managed or hybrid deployment models (see Note 1 for Gartner's initial market coverage).

Table 1: Representative Telemetry Pipeline Vendors

(Enlarged table in Appendix)

<i>Vendor</i> ↓	<i>Product Name</i> ↓	<i>Supported Deployment Models</i> ↓
Chronosphere (Calyptia)	Telemetry Pipeline	SaaS and self managed
Cribl	Stream	SaaS and self managed
Datadog	Observability Pipelines (Vector)	Observability Pipelines Worker, self-hosted options (such as, Docker, Kubernetes or a Linux server)
Edge Delta	Telemetry Pipelines	SaaS and self managed
Mezmo	Telemetry Pipeline	SaaS and self managed
Bindplane	Bindplane	SaaS and self managed
Grepr	Grepr (Intelligent Observability Data Engine)	SaaS and private cloud
Honeycomb	Honeycomb Telemetry Pipeline	SaaS
Gurukul	Data-Optimizer	SaaS and self managed
Observe AI	AI Data Pipeline	Cloud-native, hybrid, self-managed
CeTu	CeTu	Cloud-native, self-hosted and integrations
Solution Path for Optimizing Monitoring and Observability , Gartner. Monitoring and Observability for Modern Services and Infrastructure , Gartner. Focus on Telemetry Pipeline to Achieve Success in Your Observability Initiative , Gartner.		

Source: Gartner (September 2025)

Market Recommendations

Before implementing telemetry pipelines, I&O leaders must establish robust data governance policies that define:

- Log structure and formatting (including standard fields and severity levels)
- Event sorting
- Handling of unstructured and extraneous data
- Filtering requirements

- Sensitive data masking
- Metric aggregation and summarization

This foundation is crucial for effective data transformation and enrichment within the pipeline. Following on from this, I&O leaders should:

- **Consolidate and unify telemetry:** Eliminate telemetry silos by transitioning to a single, unified pipeline for all metrics, logs and traces. This approach improves team collaboration, streamlines incident response, and reduces redundant data collection and costs across the enterprise.
- **Prioritize filtering and cost optimization in telemetry:** Prioritize filtering and reduction within the telemetry pipelines to control data volume sent to costly analysis systems. Optimize the budget by routing noncritical data to lower-cost storage for compliance or less time-sensitive analysis.
- **Maximize telemetry pipeline capabilities:** Don't just use telemetry pipelines for cost reduction, leverage their full capabilities. This means transforming and enriching data with contextual information — like container or pod names, IDs or service names — and reformatting it for consistent naming. These efforts increase telemetry's value and simplify analysis for all teams.
- **Plan for telemetry pipeline integration and operations:** Thoroughly plan the telemetry pipeline's integration with existing monitoring. Develop a phased cutover plan to minimize disruption, accounting for infrastructure and skill gaps. Finally, ensure continuous training for the I&O team to maximize value.
- **Embrace vendor neutrality and open standards for telemetry:** Prioritize telemetry solutions with strong integration to industry-standard interfaces (e.g., OpenTelemetry, Splunk HEC, Syslog, Open Cybersecurity Schema Framework [OCSF]). This approach enhances flexibility, reduces vendor lock-in and streamlines data transport to various destinations, allowing easier adaptation to evolving technologies.

Acronym Key and Glossary Terms

OpTel	operational telemetry
SecTel	security telemetry
OTel	OpenTelemetry
OTLP	OpenTelemetry Protocol
MELT	metrics, events, logs and traces

Evidence

Note 1: Gartner's Initial Market Coverage

This Market Guide provides Gartner's initial coverage of the market and focuses on the market definition, rationale for the market and market dynamics.

Notes

Recommended by the Author

Some documents may not be available as part of your current Gartner subscription.

[Focus on Telemetry Pipeline to Achieve Success in Your Observability Initiative](#)

[Magic Quadrant for Observability Platforms](#)

[Assessing OpenTelemetry for Enterprise Observability](#)

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